

Different Oscillation Suppression Methods

Table 1

Suppression Method	Method in Brief	Advantages	Disadvantages
Device selection	Select devices with small package inductance	Easy to implement	Limited suppression effect
PCB layout optimisation	PCB layout to minimize the parasitic inductances and their effects.	No additional circuits required, and suppression effect is significant	Complex PCB design, and difficult to implement
External gate resistor	Adding an external gate resistor to reduce dv/dt and di/dt	Easy to implement	Increased switching loss, slow switching speed, limited suppression effect
Ferrite beads	Adding a ferrite bead, increasing high frequency damping	Easy to implement	Increased conduction loss
RC snubber	Add RC snubber by RLC equivalent model of circuit	Easy to implement	Increased switching loss, and limited suppression effect
Active gate driver	Active gate driver circuit to control gate loop impedance	Significant suppression, reduced switching losses, and maintain faster switching speeds	Complicated circuit design, and additional costs

impedance of the ferrite bead.

For suppressing the parasitic ringing effectively, following conditions must be satisfied:

(i) The damping coefficient m must be greater than 1

(ii) The above condition needs to be met at the oscillation frequency, and the ferrite bead impedance at the oscillation frequency should be mainly resistive

(iii) The DC power loss of the ferrite bead should be low

Talking about RC snubbers, they are a combination of impedances used to suppress the rapid rise in voltage across a switching device. The voltage across a parasitic capacitor cannot change instantaneously, so a decreasing transient current will flow through it for a fraction of a second, allowing the voltage across the switch to increase more slowly when the switch is opened.

At the gate of the GaN transistor, second-order RLC model can be used for parasitic ringing during the turn-on and turn-off transients of the device. RC snubbers can be designed to suppress this parasitic ringing. The snubber elements can be valued as follows:

$$R_{snb}(s) = (1/2\zeta)(\sqrt{L_p/C_p})$$

$$C_{snb}(s) = 1/2\pi R_{snb} f_r$$

where ζ is the damping ratio, L_p is the parasitic inductance, C_p is the parasitic capacitance of the device, and f_r is the oscillation frequency.

PCB layout practices

Parasitic inductances and capacitances are introduced by device packaging and PCB layout. By optimising the PCB layout, we can suppress the effects of parasitic elements and as a

result reduce oscillations and ringing. PCB layout techniques can minimise the power and drive loop inductance, and the optimal double-sided layout can reduce the power loop inductance significantly, suppressing the oscillations effectively.

Reducing parasitics via PCB layout has the benefit that it does not require any additional auxiliary circuit. But with this, the layout may become complex, and these techniques are rather hard to implement.

Impact of PCB layout parasitics

To intuitively understand how each major parasitic element in PCB layout affects the performance of the circuit design, we model the circuit with device and parasitic inductances and capacitances.

The power loop inductance (L_p/L_s) is one of the major parasitic inductances affecting the reliability of the system. Due to this inductance, overshoots and ringing occurs during turn-on and turn-off transients. To reduce these effects, the power loop length must be minimised and the decoupling capacitors must be placed close to the power device.

Another major inductance is common-source inductance (L_{cs}) or CSI. This inductance slows down the switching speed and gives rise to gate ringing and oscillations. It is critical from a stability viewpoint. For minimising the effect of this parasitic inductance, Kelvin connection must be implemented to source.

From the driver and gate interface, we get gate-loop inductance which causes undershoots, overshoots, and

false turn-on. For suppressing its effects, the gate driver must be placed close to the GaN FET to minimise gate loop.

Talking about parasitic capacitance, gate-drain coupling capacitance is by far the major parasitic capacitor. This capacitor limits the switching speed and participates in sustained oscillations. To minimise its effect, overlapping between drain and gate drive copper pour or track must be avoided.

Recommended practices

Following are some of the recommended PCB layout practices summarised for optimal performance:

1. Place the gate driver close to the GaN FET and minimise the gate drive loop area and length for optimising gate drive loop

2. Increase RG or use a ferrite bead on the gate if gate oscillation is observed

3. Use a tight layout to minimise power loop length and place the decoupling capacitor as close as possible

4. Minimise noise coupling due to PCB parasitic capacitance

5. Minimise overlapping between drain side power connection and gate drive signal track

6. Design for low thermal resistance using thermal vias and copper pours

7. When paralleling, implement dual side placement with optimum layout. Use individual gate resistors and star connection for gate drive.

In this article, we came across various oscillation and overshoot suppression techniques. Table 1 shows comparison among them. **EFY**